

Probability basics: distributions, conditioning, and expectation

We assume the definitions of probability, random variables, CDFs, densities, and expectation. The main new idea is that information changes a distribution, and therefore changes the averages we compute from it. Throughout, assume that every random quantity we average is integrable: its expected absolute value is finite.

1. Joint, marginal, and conditional distributions

Suppose X and Y each take values in $\{0, 1\}$, with the following *joint distribution*. Each interior entry is the probability that both indicated values occur.

	$Y = 0$	$Y = 1$	Total
$X = 0$	3/8	1/8	1/2
$X = 1$	1/8	3/8	1/2
Total	1/2	1/2	1

The row and column totals give the *marginal distributions* of X and Y , respectively, by summing over the other variable. Both are fair bits, but their values agree with probability $3/4$; the marginals alone do not determine the joint distribution.

Conditioning restricts and renormalizes

If we learn that $X = 0$, only the first row remains possible. Its entries sum to $1/2$, so divide each entry by $1/2$ to obtain probabilities summing to 1:

$$\Pr[Y = 0 \mid X = 0] = \frac{3}{4}, \quad \Pr[Y = 1 \mid X = 0] = \frac{1}{4}.$$

This is the *conditional distribution* of Y given $X = 0$. The general rule, whenever $\Pr[X = x] > 0$, is

$$\Pr[Y = y \mid X = x] = \frac{\Pr[X = x, Y = y]}{\Pr[X = x]}.$$

Conditioning on $X = 1$ instead gives probabilities $1/4$ and $3/4$. The variable Y is the same; the information used to describe it has changed.

Reassemble the original distribution

We recover the marginal probability of $Y = y$ by averaging its conditional probabilities, weighted by the probabilities of the conditioning cases:

$$\Pr[Y = y] = \sum_x \Pr[X = x] \Pr[Y = y \mid X = x].$$

For the table, $\Pr[Y = 1] = (1/2)(1/4) + (1/2)(3/4) = 1/2$. This is the *law of total probability*.

Bayes' rule expresses the same joint probability in the other order:

$$\Pr[X = x \mid Y = y] = \frac{\Pr[Y = y \mid X = x]\Pr[X = x]}{\Pr[Y = y]}.$$

The denominator must be positive. Conditional probabilities in opposite directions are generally different; equality in this particular symmetric table is special.

2. Conditional expectation is an average under a conditional law

Return to the table. Since Y is either 0 or 1, its expectation is the probability that it is 1. Without any information, $\mathbb{E}[Y] = 1/2$. With $X = 0$ known, we instead average using the conditional probabilities:

$$\mathbb{E}[Y \mid X = 0] = 0 \cdot \frac{3}{4} + 1 \cdot \frac{1}{4} = \frac{1}{4}.$$

Similarly, $\mathbb{E}[Y \mid X = 1] = 3/4$. More generally, for a discrete random variable Y ,

$$\mathbb{E}[Y \mid X = x] = \sum_y y \Pr[Y = y \mid X = x].$$

Conditioning on an event

We can also condition on any event A with positive probability. The rule is the same: restrict to A , renormalize, and average under that conditional distribution. For a discrete Y ,

$$\Pr[Y = y \mid A] = \frac{\Pr[\{Y = y\} \cap A]}{\Pr[A]}, \quad \mathbb{E}[Y \mid A] = \sum_y y \Pr[Y = y \mid A].$$

For example, let Y be a fair die roll and A the event that the roll is even. Given A , each of 2, 4, 6 has probability 1/3, so $\mathbb{E}[Y \mid A] = (2 + 4 + 6)/3 = 4$.

An *indicator* $\mathbf{1}\{A\}$ equals 1 on A and 0 otherwise. Multiplying by it keeps the value of Y on A and replaces it by zero elsewhere. Thus

$$\mathbb{E}[Y \mathbf{1}\{A\}] = \Pr[A] \mathbb{E}[Y \mid A].$$

For the die, both sides equal $(2 + 4 + 6)/6 = 2$. The conditional mean is 4, while this unconditional average is 2: the latter also includes the zero contributions outside A .

A number versus a random variable

For each fixed x , let $m(x) = \mathbb{E}[Y \mid X = x]$. This is a function of the information supplied. In our example,

$$m(0) = \frac{1}{4}, \quad m(1) = \frac{3}{4}, \quad m(x) = \frac{1}{4} + \frac{x}{2} \quad (x \in \{0, 1\}).$$

The notation $\mathbb{E}[Y \mid X]$ means the random variable $m(X)$. Before X is observed, its value is uncertain; after $X = x$ is observed, it takes the value $m(x)$. It is an average of possible values of Y , not the actual value of Y .

The law of total expectation

If we average the conditional averages over X , we recover the original expectation:

$$\boxed{\mathbb{E}[\mathbb{E}[Y \mid X]] = \mathbb{E}[Y].}$$

This is also called the *tower property*. In the example, it gives

$$\mathbb{E}[m(X)] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{3}{4} = \frac{1}{2} = \mathbb{E}[Y].$$

For discrete variables, the proof is the law of total probability inside a weighted sum:

$$\begin{aligned}\mathbb{E}[\mathbb{E}[Y \mid X]] &= \sum_x \Pr[X = x] \sum_y y \Pr[Y = y \mid X = x] \\ &= \sum_y y \sum_x \Pr[X = x, Y = y] = \sum_y y \Pr[Y = y] = \mathbb{E}[Y].\end{aligned}$$

What becomes fixed inside a conditional average?

Once $X = x$ is known, a function of X is a constant. Thus

$$\mathbb{E}[X + Y \mid X = x] = x + \mathbb{E}[Y \mid X = x].$$

In random-variable notation, $\mathbb{E}[X + Y \mid X] = X + \mathbb{E}[Y \mid X]$. More generally, conditional expectation is linear, and a known factor can be taken outside:

$$\mathbb{E}[a(X)Y \mid X] = a(X)\mathbb{E}[Y \mid X],$$

No independence assumption is required: the factor is known because we conditioned on X .

3. Conditioning with continuous random variables

For a variable with density f_X , the probability of an interval is its density integrated over that interval. A density value $f_X(x)$ is not the probability of the point x ; that point has probability zero. In particular, the discrete conditional-probability formula cannot be used by dividing by $\Pr[X = x]$.

Joint densities and conditional densities

A joint density $f_{X,Y}$ describes probability over pairs of values. For intervals A, B ,

$$\Pr[X \in A, Y \in B] = \int_A \int_B f_{X,Y}(x, y) dy dx.$$

To get the marginal density of X , integrate over all possible values of Y :

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy.$$

Fix x with $f_X(x) > 0$. The function $y \mapsto f_{X,Y}(x, y)$ has total integral $f_X(x)$. Dividing by that total produces a density with integral 1:

$$f_{Y|X}(y | x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

This is the continuous analogue of renormalizing a row of the joint probability table. It defines conditioning on the continuous value through densities, not by a ratio of point probabilities.

A worked example

Let $f_{X,Y}(x, y) = x + y$ for $0 \leq x, y \leq 1$, and zero elsewhere. It is nonnegative, and its integral over the square is 1. For $0 \leq x \leq 1$,

$$f_X(x) = \int_0^1 (x + y) dy = x + \frac{1}{2}, \quad f_{Y|X}(y | x) = \frac{x + y}{x + 1/2} \quad (0 \leq y \leq 1).$$

For instance, when $x = 0$, the conditional density is $2y$, which places more weight near 1 than near 0. A conditional mean uses this density in the usual expectation formula:

$$m(x) = \mathbb{E}[Y | X = x] = \int_0^1 y \frac{x + y}{x + 1/2} dy = \frac{x/2 + 1/3}{x + 1/2}.$$

In particular, $m(0) = 2/3$, while $m(1) = 5/9$. The conditional distribution, and its average, depend on the observed value.

The tower property now averages using the marginal density of X :

$$\mathbb{E}[Y] = \int_0^1 m(x) f_X(x) dx = \int_0^1 \left(\frac{x}{2} + \frac{1}{3} \right) dx = \frac{7}{12}.$$

To check directly, the marginal density is $f_Y(y) = y + 1/2$ on $[0, 1]$, giving $\mathbb{E}[Y] = \int_0^1 y(y + 1/2) dy = 1/3 + 1/4 = 7/12$.

4. Independence and conditional independence

Independence means that learning one variable does not change the law of the other. For discrete variables, it is equivalent to

$$\Pr[X = x, Y = y] = \Pr[X = x] \Pr[Y = y] \quad \text{for all } x, y.$$

When densities exist, the analogous condition is $f_{X,Y}(x, y) = f_X(x) f_Y(y)$ almost everywhere. Dividing by the marginal then gives $f_{Y|X}(y | x) = f_Y(y)$ wherever the ratio is defined. In particular, under independence,

$$\mathbb{E}[Y | X] = \mathbb{E}[Y].$$

Without independence, averaging over Y while fixing $X = x$ generally requires the conditional law of Y , not its marginal law.

For independent variables, functions of the separate variables remain independent. Hence $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$. Linearity, by contrast, never requires independence: $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$.

A function of independent variables

Suppose we want the conditional average of a quantity $g(X, Y)$ that depends on both variables. Under independence, fixing $X = x$ leaves the distribution of Y unchanged. We substitute x into the function and average over Y :

$$\mathbb{E}[g(X, Y) \mid X = x] = \mathbb{E}[g(x, Y)].$$

For discrete Y , the right side is $\sum_y g(x, y)\Pr[Y = y]$; with a density it is $\int g(x, y)f_Y(y) dy$.

For example, let X, Y be independent uniform draws from $[0, 1]$ and let $g(X, Y) = XY + Y^2$. For fixed x ,

$$\mathbb{E}[XY + Y^2 \mid X = x] = \int_0^1 (xy + y^2) dy = \frac{x}{2} + \frac{1}{3}.$$

The answer still depends on x : independence does not remove the function's dependence on the variable we fixed. Without independence, the same calculation would use the conditional distribution of Y given $X = x$, as in the preceding section.

Independence after information is supplied

Variables X and Y are *conditionally independent given H* if their conditional joint distribution factors within each conditioning case. For discrete variables,

$$\Pr[X = x, Y = y \mid H = h] = \Pr[X = x \mid H = h]\Pr[Y = y \mid H = h]$$

for all h with positive probability. Equivalently, once H is known, learning X gives no additional information about Y .

Example: two tosses of a randomly selected coin. Choose one of two coins with equal probability. The first has head probability $1/4$, and the second has head probability $3/4$. Let $H \in \{1, 2\}$ identify the selected coin, and let X, Y indicate heads on two tosses. Conditional on the chosen coin, the tosses are independent. Without knowing the coin,

$$\Pr[X = 1] = \Pr[Y = 1] = \frac{1}{2}, \quad \Pr[X = 1, Y = 1] = \frac{1}{2} \left(\frac{1}{4}\right)^2 + \frac{1}{2} \left(\frac{3}{4}\right)^2 = \frac{5}{16}.$$

Since $5/16 \neq (1/2)(1/2)$, the tosses are not unconditionally independent. They share uncertainty about which coin was selected.

The unconditional distribution is a *mixture*: select a case, then draw from its distribution. Its probabilities are weighted averages of the component probabilities, by the law of total probability.

After observing $X = 1$, Bayes' rule updates the probability of the second coin:

$$\Pr[H = 2 \mid X = 1] = \frac{\Pr[X = 1 \mid H = 2]\Pr[H = 2]}{\Pr[X = 1]} = \frac{(3/4)(1/2)}{1/2} = \frac{3}{4}.$$

The posterior weights on the two coins are now $1/4$ and $3/4$. The probabilities of heads given each coin remain $1/4$ and $3/4$. These two pairs coincide only because of our symmetric choice of biases and equal prior weights. Averaging with the updated weights gives

$$\Pr[Y = 1 \mid X = 1] = \frac{1}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{3}{4} = \frac{5}{8}.$$

In general, conditional independence permits the calculation

$$\Pr[Y = y \mid X = x] = \sum_h \Pr[H = h \mid X = x] \Pr[Y = y \mid H = h].$$

The first factor records what we now know about H ; the second describes Y within that case.

The converse also fails: conditioning can destroy independence. If X, Y are initially independent fair bits, then conditional on $X + Y = 1$, learning X determines $Y = 1 - X$.

5. Functions of random variables

A function of a random variable is another random variable. Two different questions are useful: what is its distribution, and what is its expectation? To find the expectation, we can work directly with the original distribution:

$$\mathbb{E}[g(X)] = \sum_x g(x) \Pr[X = x] \quad \text{or} \quad \mathbb{E}[g(X)] = \int g(x) f_X(x) dx.$$

The second formula applies when X has a density. This is sometimes called the *law of the unconscious statistician*. For $X \sim U[0, 1]$, it gives $\mathbb{E}[X^2] = \int_0^1 x^2 dx = 1/3$.

To find the distribution of $Z = g(X)$, use its defining event: $F_Z(z) = \Pr[g(X) \leq z]$. For the same uniform X and $Z = X^2$, nonnegativity of X gives

$$F_Z(z) = \begin{cases} 0 & z < 0, \\ \Pr[X \leq \sqrt{z}] = \sqrt{z} & 0 \leq z \leq 1, \\ 1 & z > 1. \end{cases}$$

Thus Z is not uniform. Its density is $1/(2\sqrt{z})$ on $(0, 1)$, and integrating z against that density also gives $\mathbb{E}[Z] = 1/3$.

Nonlinear functions do not commute with expectation

For the uniform example, $(\mathbb{E}[X])^2 = 1/4$, whereas $\mathbb{E}[X^2] = 1/3$. Linearity lets us move an affine function through expectation; it does not let us move an arbitrary function through it.

A function g on an interval is *convex* if

$$g(\alpha x + (1 - \alpha)y) \leq \alpha g(x) + (1 - \alpha)g(y) \quad (0 \leq \alpha \leq 1).$$

The graph lies below each chord. *Concavity* reverses the inequality. *Strict convexity* makes the inequality strict for distinct x, y and $0 < \alpha < 1$.

Jensen's inequality. For convex g , when X lies in its domain,

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

For strictly convex g , equality holds exactly when X is constant with probability 1. Concave functions reverse the inequality. In particular, x^2 is convex, while $\log x$ is concave on the positive reals.

For differentiable convex g and $m = \mathbb{E}[X]$ in the interior of its domain, the proof comes from the tangent line: $g(x) \geq g(m) + g'(m)(x - m)$. Taking expectations makes the last term zero. Strict convexity makes the tangent inequality strict whenever $x \neq m$, giving the equality condition. The general version uses a supporting line and is stated here without proof.

A few notation conventions

For the indicator defined earlier, $\mathbb{E}[\mathbf{1}\{A\}] = \Pr[A]$. *Almost surely* means with probability 1. A distribution is *atomless* if every individual point has probability zero; therefore $\Pr[X < t] = \Pr[X \leq t]$ for any fixed t , and $\Pr[X \geq t] = 1 - F_X(t)$. The notation $\int g dF$ means expectation under the law F , using a sum or density integral when available. A subscript such as $\mathbb{E}_X$ identifies which randomness is being averaged.